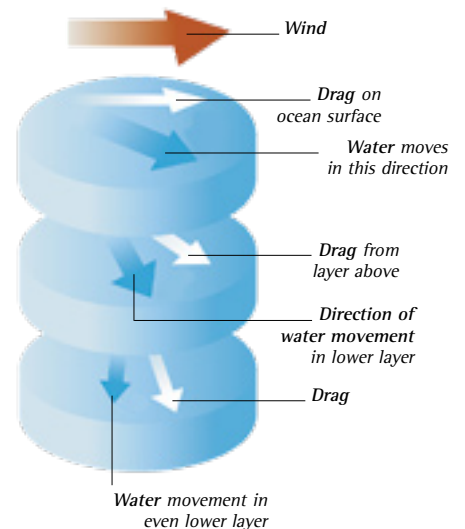


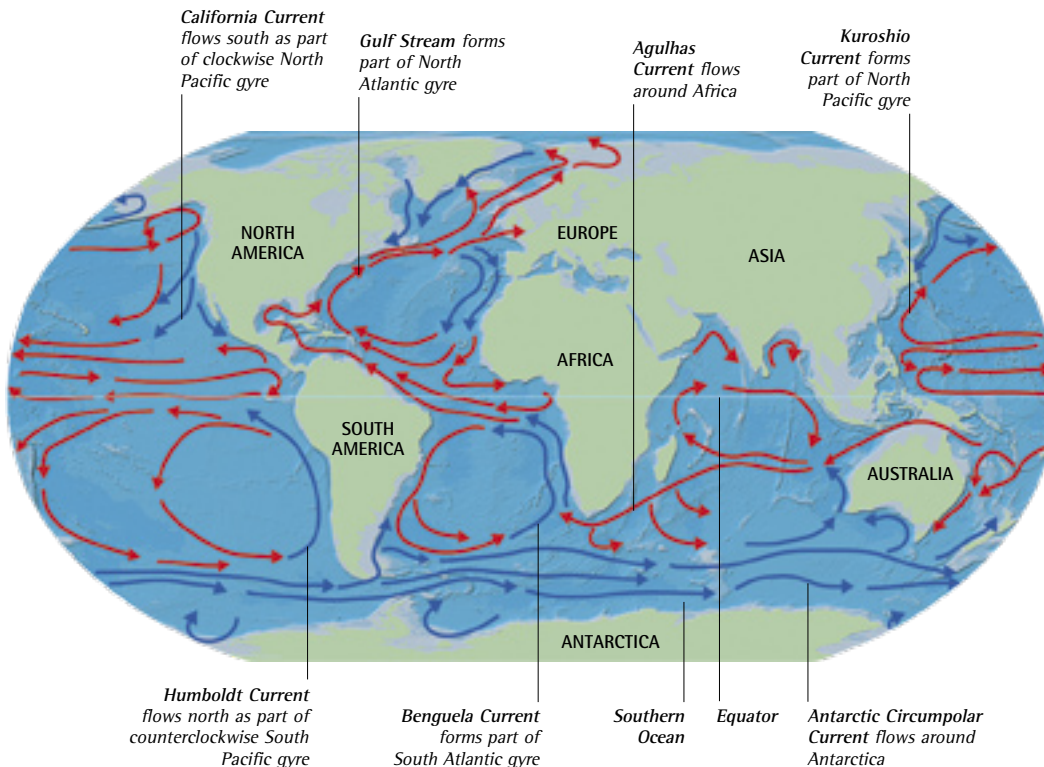
SURFACE CURRENTS

The winds that sweep over the world's oceans drag the surface waters along with them, helping to drive the surface currents that flow around the oceans. The most influential of these winds are the prevailing winds generated by global air circulation. Deflected by the Earth's spin, they swerve toward the west in the tropics and toward the east in the temperate zones, driving the water in similar directions. The resulting surface currents form huge, swirling gyres that redistribute warm and cold water around the globe.



▲ OCEAN RIVERS

Surface currents flow like immense rivers through surrounding ocean waters. The Gulf Stream, for example, transports water northeast through the North Atlantic at some 1,766 million cubic feet (50 million cubic meters) a second—thousands of times the flow rate of the Amazon River in South America. This view from space shows the boundary between the fast-moving Gulf Stream at the bottom of the picture, and the coastal waters of North America at the top.



▲ THE EKMAN SPIRAL

Winds blowing toward the east don't simply push water eastward. The spinning Earth effect that makes the wind swerve off-course also makes moving water veer right to the north of the equator, or left to the south. The moving water drags deeper water with it, and this also swerves right or left. The result is a current shear that increases with depth, called the Ekman spiral after its discoverer, Vagn Walfrid Ekman (1874–1954). The total effect is that surface currents flow at about 45 degrees to the wind direction.

◀ SWIRLING GYRES

The combination of prevailing winds and the Ekman effect carries surface water westward near the equator, and toward the east in the temperate regions. In the Southern Ocean, the eastward flow continues around Antarctica, but elsewhere the continents force the currents to form ocean-scale circulations called gyres. These flow in opposite directions north and south of the equator, while offshoots of the main gyres flow into the Arctic Ocean and around southern Africa.